

Ex. No: 04 - Analyze Email Headers and Detect Email Spoofing using MHA

AIM:

To inspect raw email headers, trace server routing hops, and analyze authentication protocols (SPF, DKIM, DMARC) using Mail Header Analyzer (MHA) to identify email spoofing and phishing attempts.

PROCEDURE

1. **Extract Email Header Data:** Open the target email in Gmail, click **More** (three vertical dots), and select **Show original**.
2. **Copy Raw Header Text:** Highlight and copy the full header text, including all metadata fields.
3. **Identify Key Header Metadata Fields:** Locate the From, To, Date, Subject, Return-Path, Received, and Message-ID fields.
4. **Analyze Server Hops in 'Received' Lines:** Trace the path of the email starting from the bottom-most Received: line up to the top.
5. **Validate Email Authentication Protocols:** Check the status of SPF, DKIM, and DMARC.
6. **Automate Parsing via MHA:** Paste the copied raw header into **Google Admin Toolbox Messageheader**.
7. **Detect Spoofing Anomalies:** Look for domain mismatches, authentication failures, or unexpected routing delays.

DOCUMENTATION OF FINDINGS

1. Initial Header Extraction (Gmail)

- **From:** SBI <sbi@communications.sbi.co.in>
- **To:** <sanjeevk23185@gmail.com>
- **Subject:** Your card, your rules: Customize your transaction limits on YONO SBI
- **Message-ID:** wvumeo17872509055276640@communications.sbi.co.in

Original message	
Message ID	<wvumeo17872509055276640@communications.sbi.co.in>
Created on:	21 August 2026 at 22:11 (Delivered after 0 seconds)
From:	SBI <sbi@communications.sbi.co.in> Using MIME::Lite 3.033 (F2.85; T2.24; A2.21; B3.16; Q3.16)
To:	sanjeevk23185@gmail.com
Subject:	Your card, your rules: Customize your transaction limits on YONO SBI
SPF:	PASS with IP 175.158.67.76 Learn more
<td>'PASS' with domain communications.sbi.co.in Learn more</td>	'PASS' with domain communications.sbi.co.in Learn more
DMARC:	'PASS' Learn more
Download original Copy to clipboard	
Delivered-To: sanjeevk23185@gmail.com Received: by 2002:a05:f6d:cce:b6:120-45df:c6f4 with SMTP id v4lcpic188BAArcg; Fri, 21 Aug 2026 09:41:42 -0700 (PDT) X-Received: by 2002:a05:f6d21:5581:b0:3c4:511e:26ab with SMTP id adffeje73aafo8_3cd2fdb0915mr15387579637.3.1787385081727; Fri, 21 August 2026 09:41:41 -0700 (PDT) ARC-Seal: i=1; ar-sra-sha256; t=-1787330501; cv=None; d-google.com; s=sarc-20260327; b=xMqZGhqshnKhTgh/UwbAmUWbkCYjdShbaqpHrPFsn3jGuGUJcdKDeezAYCVmS0 CeeAU 3Juak6R3PDURxVUUQOTTHdlRFjlRHlvgrfaPaTBesrLUDHKMYeyeYeghtgdihy hTtGLPV0IzDPRVLztNIVysivoumdsgOstqUbmsGOULtaJ+CBMFrbksxdpxnmE bHg-- ndPrVVmtU/vzgjrllTVsyovumdegsOstqUbmsGOULtaJ+CBMFrbksxdpxnmE bHg-- ARC-Message-Signature: i=1; ar-sra-sha256; c=rElaxed/rElaxed; d-google.com; s=sarc-20260327; b-feedback-id:list-unsubscribecontent-transfer-encoding:nme-version :message ;id:subject:from:date:dkim-signature:dkim-signature; bh:BfpvvLDZphnCqbqvSSl7BQMqWJBy/fASVPvtg; fh=somdslnPf/zooAhdEvexGaRDQYQ3As5G; b=dm; l=/FSJoapbyebhcLEsfFlAlMDyeOmssaulsuUZUlzllovdemosemail.iurfl	

2. MHA Routing and Authentication Analysis (Google Admin Toolbox)

- **Authentication Check:**

- **SPF:** PASS
- **DKIM:** PASS (Domain: communications.sbi.co.in)
- **DMARC:** PASS

- **Hop-by-Hop Analysis:**

- **Hop 1 (Origin):** From d67-76.smtp-out.in.alerts.sbi.co.in. to Google's receiving server (mx.google.com) via ESMTPS.
- **Hop 2 & 3 (Internal Routing):** Internal Google SMTP routing to the final inbox.

- **Delivery Time:** The total transit time from the sender to the inbox was exactly 1 second, indicating no suspicious routing delays or rerouting through unauthorized third-party servers.

Google Admin Toolbox Messageheader							Help
MessageID	wvumeo17872509055276640@communications.sbi.co.in						
Created at:	8/21/2026, 10:11:41 PM GMT+5:30 (Delivered after 1 sec)						
From:	SBI <sbi@communications.sbi.co.in> Using MIME-Lite 3.033 (F2.85; T2.24; A2.21; B3.16; Q3.16)						
To:	<sanjeevk23185@gmail.com>						
Subject:	Your card, your rules: Customize your transaction limits on YONO SBI						
DKIM:	pass with domain communications.sbi.co.in Learn more						
#	Delay	From *		To *	Protocol	Time received	
0		d67-76.smtp-out.in.alerts.sbi.co.in.	→	[Google] mx.google.com	ESMTPS	8/21/2026, 10:11:41 PM GMT+5:30	
1			→	[Google] 2002:a05:6a21:1583:b0:3c4:511e:26ab	SMTP	8/21/2026, 10:11:41 PM GMT+5:30	
2	1 sec		→	[Google] 2002:a05:6f02:ea9:b0:129:45df:65f4	SMTP	8/21/2026, 10:11:42 PM GMT+5:30	

CONCLUSION: No anomalies, time discrepancies, or domain mismatches were found. The From domain matches the cryptographic Message-ID domain. All authentication protocols (SPF, DKIM, DMARC) passed successfully. The email is a legitimate, authorized communication and is **not spoofed**.